

Shi Boao

github.com/bowieshi
<http://www.bowieshi.com>
aoibosh@connect.hku.hk

EDUCATION

The University of Hong Kong

Bachelor of Engineering, major in Computer Science, minor in Math&Stat

Sept. 2022 - Present

University of Illinois at Urbana-Champaign (Semester exchange)

The Grainger College of Engineering

GPA: 4.12/4.3 Rank: 2/140+

2025 Spring

Hangzhou No.2 High School of Zhejiang Province

Sept. 2019 - June. 2022

RELEVANT COURSEWORK

Grades: 14 A+ out of 17 major courses

Courses: Computer Graphics: A+; Linear algebra A+; Probability & Statistics I: A+; Calculus and ordinary differential equations: A+; Operating System: A+; Object-oriented Programming: A+; Discrete Math: A+; Data Structures and Algorithms: A+; Quantum information and computation: A+; Principles of programming languages: A+; etc.

SKILLS

Communication Languages: Mandarin: Native; English: Fluent (TOEFL iBT: **104** (with sub-scores 24+))

Main Programming Languages: Python $\approx$ C/C++ > Java

Full Stack Developing: HTML, CSS, JavaScripts(JQuery, ReactJS, NodeJS), MySQL/MongoDB, PHP

Packages: Pytorch, ROS1, ROS2, IsaacGym, OpenCV, colmap, 3DGS

Software: Docker, Blender, MeshLab, Unity

Hardware: Orbbec femto RGBD camera, Intel NUC

PUBLICATIONS

EvoR: Evolving Retrieval for Code Generation

*Hongjin Su, Shuyang Jiang, Yuhang Lai, Haoyuan Wu, **Boao Shi**, Che Liu, Qian Liu, Tao Yu*

Accepted as The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP2024), Findings

INTERNSHIP

Research Intern in Infobodied Lab | *Robotics, Computer Vision, Computer Graphics*

May. 2024 – Now

- Research in Prof. Yanchao Yang's lab in HKU Musketeers Foundation Institute of Data Science, focusing on Embodied AI. I am leading a research project about humanoid scene interaction generation, which is prepared to submit to ICCV2025
- Understanding the state of the art in embodied motion generation of humanoid robots, including motion capture, motion synthesis, and motion control
- Build up an automatic human whole body motion data collection system, which extract demonstration data from real human video and reconstruct real scenes into their digital twins; reconstruct interacted objects and track their 6D pose; collecting the largest dataset to date for human-scene/object interaction
- Using deep learning to train a text-driven scene-conditioned human motion generation model, which can be used as high-level motion guide for low-level humanoid robots control

Research Assistant in XLang Lab | *Natural language processing, LLM*

Dec. 2023 – Mar. 2024

- Participated in a research project about Information Retrieval and Code Generation
- Built a benchmark for testing information retrieval model reasoning ability, including data collection, data preprocessing
- Curated a dataset about long-tail programming Languages, testing LLM generalization ability when aided with retrieval argumentation generation
- Outputted a research paper "EvoR: Evolving Retrieval for Code Generation"

PROJECTS

Robot auto-aiming system <i>Robotics, Computer Vision, Control Theory, Sensors</i>	May. 2023 – Apr. 2024
- Using traditional algorithm(OpenCV) or Yolo model to recognize lights on armors - Trained a shallow layer neural network for fast recognizing number on armors - Use PnP algorithm to estimate pose of target armor in 3D space and control robot to shoot through ROS - Using Extended Kalman Filter algorithm to predict, denoise, and track target's moving and spinning status	

EXPERIENCE & TEACHING

HKU Innovation Wing RoboMaster Team <i>Member</i>	Sept. 2022 - Apr. 2023
Leader of computer vision group, develop first generation of robot auto-aiming system Won Eighth-ranked team in both 3V3 and Infantry Match in RoboMaster University League (RMUL) held by DJI	
HKU ENGG1330 (programming I) <i>Teaching Assistant</i>	Sept. 2023 – Dec. 2023
As student teaching assistant for 2-hours tutorial every week	
HKU COMP2396 (Object-oriented Programming and Java) <i>Teaching Assistant</i>	Sept. 2024 – Dec. 2024
As student teaching assistant for tutorial questions answering	

COMPETITON AWARDS

Huawei Coding Challenge	March 2023
Rank: 25/311, Third Prize	
2022 International Collegiate Programming Contest Asia East Continent Final Contest	March 2023
Bronze Medal	
2022 International Collegiate Programming Contest Hong Kong Regional Contest	Jan. 2023
Rank: 19, Silver Medal	
National Olympiad in Informatics in Provinces(NOIP) (Zhejiang Province)	2018 - 2020
First Prize × 3 times	

TALKS

HKU CSSAUD Senior Academic Sharing Session (Engineering)	Sept. 2024
Sharing study experience in computer science, and how to begin research in AI. Record & PPT	
RoboMaster Team Training Workshop - Introduction to Linux System (Algorithm group)	Sept. 2023
Introduction to Linux System, system overview, basic command line usage, remote server usage	

SCHOLARSHIP & AWARDS

Hong Kong University Alumni Prize	Oct. 2024
Academic-merit-based Award, The University of Hong Kong (1 awardee in Faculty of Engineering)	
Talent Development Scholarship	June. 2024
HK\$10,000, Recognize talent in scope of Innovation, science and technology, The Hong Kong Government	
Dean's Honours List	Mar. 2024
Academic-merit-based Award for Engineering students in HKU (Top 10%), The University of Hong Kong	
Young Tsun Dart Scholarship	Nov.2023
HK\$15,000, Academic-merit-based Scholarship for Engineering students in HKU (Top 2 in Faculty of Engineering)	
China Soong Ching-ling Foundation Zhiyuan Scholarship	2022-2026
¥80,000 annually, Merit-based Scholarship for students pursuing Bachelor Degree in HK (~ 30 awardees nation-wide)	